

Program 4

Integrate git with docker for source-controlled Application builds

Program Structure:

program4

----->Dockerfile

----->index.js

----->package.json

----->package-lock.json

step 1: Create the Dockerfile without any extension and the following content

nano Dockerfile

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/git
GNU nano 6.2 Dockerfile
#step1 Build Stage
FROM node:18 As build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node","index.js"]
```

Step 2: Create the index.js file with following code

nano index.js

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/git
GNU nano 6.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Docker + Git Integration Program!");
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

step 3: Create file package .json by executing the command
npm init-y

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/git
GNU nano 6.2 package.json
{
  "name": "git",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  }
}
```

step 4: Initialize Git Repository locally
git config - - global user.name " "
git config - - global user.email " "
git init
git add .
git commit -m "Initial commit"

--object-format <hash>
specify the hash algorithm to use

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git init
git add .
git commit -m "Initial commit"
git branch -M main
Reinitialized existing Git repository in /home/1RV24MC073_pavithra/git/.git/
On branch main
nothing to commit, working tree clean
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git remote add origin https://github.com/<Pavithra282>/<hello>.git
bash: Pavithra282: No such file or directory
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git remote add origin https://github.com/Pavithra282/hello.git
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git push -u origin main
Username for 'https://github.com': Pavithra282
Password for 'https://Pavithra282@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Pavithra282/hello.git/'
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ █
```

step 5: Link to remote repository

```
Activities Terminal Nov 19 08:29 1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/git

-t, --track <branch> branch(es) to track
-m, --master <branch> master branch
--mirror[=(push|fetch)] set up remote as a mirror to push to or fetch from

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git init git add . git commit -m "Initial commit" git branch -M main
error: unknown switch 'm'
usage: git init [-q | --quiet] [--bare] [--template=<template-directory>] [--shared[=<permissions>]] [<directory>]

--template <template-directory>
    directory from which templates will be used
--bare
    create a bare repository
--shared[=<permissions>]
    specify that the git repository is to be shared amongst several users
-q, --quiet
    be quiet
--separate-git-dir <gitdir>
    separate git dir from working tree
-b, --initial-branch <name>
    override the name of the initial branch
--object-format <hash>
    specify the hash algorithm to use

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git init
git add .
git commit -m "Initial commit"
git branch -M main
Reinitialized existing Git repository in /home/1RV24MC073_pavithra/git/.git/
On branch main
nothing to commit, working tree clean
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git remote add origin https://github.com/<Pavithra282>/<hello>.git
bash: Pavithra282: No such file or directory
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git remote add origin https://github.com/Pavithra282/hello.git
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git push -u origin main
Username for 'https://github.com': Pavithra282
Password for 'https://Pavithra282@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Pavithra282/hello.git/'
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git remote -v
origin https://github.com/Pavithra282/hello.git (fetch)
origin https://github.com/Pavithra282/hello.git (push)
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git push -u origin main
Username for 'https://github.com': Pavithra282
Password for 'https://Pavithra282@github.com':
Enumerating objects: 726, done.
Counting objects: 100% (726/726), done.
Delta compression using up to 4 threads
Compressing objects: 100% (671/671), done.
Writing objects: 100% (726/726), 732.51 KiB | 4.52 MiB/s, done.
Total 726 (delta 146), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (146/146), done.
To https://github.com/Pavithra282/hello.git
* [new branch] main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$
```

step 6: push the code to the github

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$ git push -u origin main
Username for 'https://github.com': Pavithra282
Password for 'https://Pavithra282@github.com':
Enumerating objects: 726, done.
Counting objects: 100% (726/726), done.
Delta compression using up to 4 threads
Compressing objects: 100% (671/671), done.
Writing objects: 100% (726/726), 732.51 KiB | 4.52 MiB/s, done.
Total 726 (delta 146), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (146/146), done.
To https://github.com/Pavithra282/hello.git
* [new branch] main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/git$
```



out put

The screenshot shows a GitHub repository page for a user named Pavithra282. The repository is named 'from' and is public. The page displays the repository's main branch, 'main', with 1 branch and 0 tags. A search bar is present with the text 'Go to file'. Below the repository name, there is a table of files and folders, all of which were committed initially 2 months ago. The files listed are 'node_modules', 'Dockerfile', 'index.js', 'package-lock.json', and 'package.json'. A 'README' file is also listed but is currently collapsed. The repository has a commit history of 1 commit, with the commit hash 'a34cc95' and a commit message 'Initial commit'.

github.com/Pavithra282/from

Pavithra282 / from

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

from (Public) Pin Watch

main 1 Branch 0 Tags Go to file Add file <> Code

Pavithra282 Initial commit a34cc95 · 2 months ago 1 Commit

node_modules	Initial commit	2 months ago
Dockerfile	Initial commit	2 months ago
index.js	Initial commit	2 months ago
package-lock.json	Initial commit	2 months ago
package.json	Initial commit	2 months ago

README